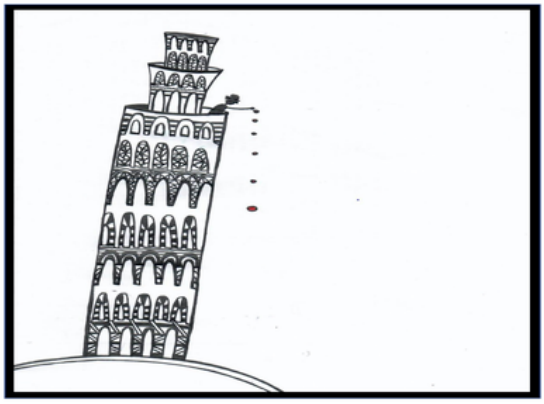


CONSERVATION OF MECHANICAL ENERGY DURING FREE FALL

Author(s): Shirish Sant, Neha Deshpande, Sugat Dabholkar

Subject: Physics, Self-directed

Time: 4 class periods (30 min each)

Level: High School / Freshmen Physics

Version: v2

OVERVIEW

This lesson is about the conservation of mechanical energy during the free fall of an object. Students will explore a computational model of free fall of a ball and make observations regarding changes in its motion and energy.

STANDARDS

NEXT GENERATION SCIENCE STANDARDS

- Physical Science
  - [HS-PS2] Motion and Stability: Forces and Interactions
  - [HS-PS2-1] Analyze data to support the claim that Newton's second law of motion describes the mathematical relationship among the net force on a macroscopic object, its mass, and its acceleration.
  - [HS-PS3-2] Develop and use models to illustrate that energy at the macroscopic scale can be accounted for as a combination of energy associated with the motions of particles (objects) and energy associated with the relative positions of particles (objects)

COMPUTATIONAL THINKING IN STEM

- Modeling and Simulation Practices
  - Designing Computational Models
  - Using Computational Models to Find and Test Solutions
  - Using Computational Models to Understand a Concept
  - Constructing Computational Models
  - Assessing Computational Models
- Computational Problem Solving Practices
  - Computer Programming
  - Troubleshooting and Debugging
- Data Practices

- Collecting Data

CREDITS

Lesson development - Shirish Sant, Neha Deshpande and Sugat Dabholkar

Topic idea - Shirish

Illustration - Neha

Model coding - Sugat

ACTIVITIES

1. Introduction to Computational Thinking
2. Exploring the model
3. Observations regarding motion of the ball
4. Changes in mechanical energy of the ball
5. Evaluating the model
6. Constructing the model

STUDENT DIRECTIONS AND RESOURCES

In this lesson, you will learn about -

1. Changes in distance, velocity, and acceleration of an object under free fall

2. Conservation of mechanical energy during the free fall of an object

3. Effect of mass on mechanical energy of an object

4. Free fall of an object is independent of its mass

5. How to use a computational model to investigate an object under free fall

6. How to evaluate and extend computational models

7. How to construct a computational model of an object under free fall using block-based coding

1. INTRODUCTION TO COMPUTATIONAL THINKING

The embedded resource from above can be accessed via this link (https://www.youtube.com/embed/2ACjQLXrdug)

In this video, researchers at the Indian Institute of Science Education and Research (IISER) Pune talk about how they use computational tools and methods in their work.

Contemporary scientists and mathematicians use such computational tools and methods to understand natural phenomena, solve problems and design solutions. The thinking required for this is called Computational Thinking.

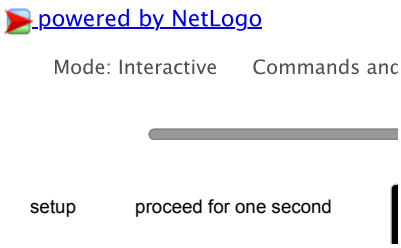
In this lesson using a computational model, we will learn about computational thinking, and motion and conservation of mechanical energy of an object during free fall.

QUESTION 1.1

Based on what you saw in the video, explain why science students need to develop computational thinking.

2. EXPLORING THE MODEL

Explore the model before you answer the questions below.



The embedded resource from above can be accessed via this link (https://ct-stem.s3.amazonaws.com/uploads/shirishsant02/2019/11/19/FREE_FALL.HTML)

QUESTION 2.1

What happens to the ball when you press 'proceed for one second' in the model?

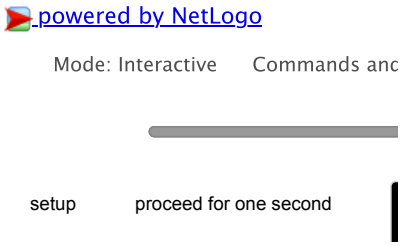
QUESTION 2.2

Write three observations about the motion of the ball.

QUESTION 2.3

Write three observations about the energy of the ball.

3. OBSERVATIONS REGARDING MOTION OF THE BALL



The embedded resource from above can be accessed via this link (https://ct-stem.s3.amazonaws.com/uploads/shirishsant02/2019/11/19/FREE_FALL.HTML)

QUESTION 3.1

What changes do you observe in distance as time progresses?

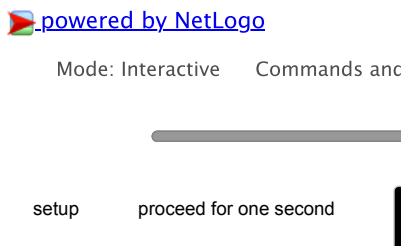
QUESTION 3.2

What changes do you observe in velocity as time progresses?

QUESTION 3.3

What changes do you observe in acceleration as time progresses?

4. CHANGES IN MECHANICAL ENERGY OF THE BALL



The embedded resource from above can be accessed via this link (https://ct-stem.s3.amazonaws.com/uploads/shirishsant02/2019/11/19/FREE_FALL.HTML)

QUESTION 4.1

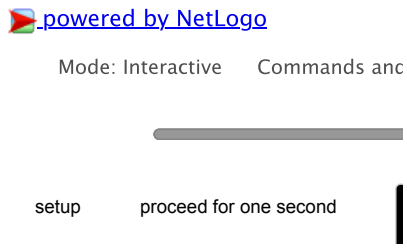
Observe what happens to kinetic energy (KE), potential energy (PE) and total energy (KE+PE) of the ball after each second it drops.

Write your observations below.

QUESTION 4.2

Observe what happens when the mass of the ball is changed. Write your observations below. Mention which aspects regarding the motion and energy of the ball change and which remain unchanged.

5. EVALUATING THE MODEL



The embedded resource from above can be accessed via this link (https://ct-stem.s3.amazonaws.com/uploads/shirishsant02/2019/11/19/FREE_FALL.HTML)

QUESTION 5.1

Do you think you learned anything new with this model? If yes, elaborate.

QUESTION 5.2

What do you think such models can be useful for?

QUESTION 5.3

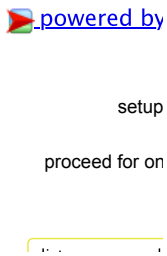
What are some limitations of this model?

QUESTION 5.4

In order to improve this model, what changes will you make?

6. CONSTRUCTING THE MODEL

On this page, you will use coding blocks to construct a model of an object under free fall.



The embedded resource from above can be accessed via this link (https://ct-stem.s3.amazonaws.com/uploads/sugat_teacher_account/2019/12/15/NETTANGO_FREE%20FALL_SCREENSHOT.HTML)

QUESTION 6.1

How will you know if your model is accurate?

QUESTION 6.2

Take screenshots of the coding blocks (setup and 'proceed for one second') in your final model and upload those here.

Upload files to the space allocated by your teacher.

QUESTION 6.3

What did you learn additionally about the motion of an object under free fall after constructing the model?

QUESTION 6.4

What is the biggest takeaway for you from this lesson?

This material was developed by the

CT-STEM Project

at Northwestern University. To inquire about using our material in your school or as part of research, please contact

ctstem@northwestern.edu

.

This work was made possible through generous support from the National Science Foundation (grants CNS-1138461, CNS-1441041, DRL-1020101, DRL-1640201 and DRL-1842374) and the Spencer Foundation (Award #201600069). Any opinions, findings, or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the funding organizations.

© 2026 CT-STEM Project. Some rights reserved.